

W Burger POS - UX/UI Architecture Plan

This document provides a comprehensive operational and architectural plan for the W Burger POS (Point of Sale) Android application. It serves as the single source of truth for the UI/UX design phase executed by Stitch.

1. APP PURPOSE SUMMARY

The W Burger POS application is the operational heart of the fast-food environment. Designed exclusively for large-screen Android terminals (landscape-first), it serves as the primary tool for cashiers and on-site managers to handle direct customer orders, process payments, validate mobile orders (via QR scan), and manage daily financial and stock closures. Its primary goal is to ensure extreme speed and accuracy during peak service hours, minimize cognitive load for staff, and enforce strict traceability for sensitive actions like drawer openings, cancellations, and daily closures.

2. TARGET USERS INSIDE THIS APP

Cashier

- Needs: Take orders instantly, scan customer QR codes quickly, apply simple notes (e.g., "sans salade"), process cash/card payments, open the cash drawer manually, and view today's tickets.
- Hidden/Protected: Applying discounts, cancelling orders, modifying stock, validating end-of-day closures. (These require Manager override).

Operational Manager (Manager)

- Needs: Everything the cashier does, plus overriding permissions (discounts, cancellations), creating purchase entries and uploading invoices, handling session closures with TPE photos, monitoring stock alerts, and performing end-of-day stock verification.
 - Hidden/Protected: Global system setup, menu creation/modification, user capability assignment. (These belong strictly in the Admin Backoffice web app, not the POS).
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3. INFORMATION ARCHITECTURE OF THE POS APP

For a fast-food POS terminal, a Split Layout with a Persistent Navigation Sidebar is the optimal architecture. Speed and context retention are paramount.

Structure Recommendation:

- Left Sidebar (Narrow, Persistent): Quick switching between core modules ([Sales](#), [QR Scan](#), [Today's Sales](#), [Stock](#), [Purchases](#), [Closures](#)). Use an icon-first approach. At the bottom: Current User profile, Setup/Utils, Logout.

- Main Content Area (Center-Left): Varies based on the selected module. In "Sales" mode, it displays the category tabs and product grid.
- Right Panel (Sticky Cart/Ticket): Specifically in "Sales" mode, the right 30-35% of the screen is permanently dedicated to the Order Ticket (Cart), showing line items, quantities, notes, totals, and the massive `Checkout/Pay` button.
- Modal Layer: Overlays used for focused, interruptive flows (Payment popup, Product modifier/note entry, Cash drawer opening reason, Closing discrepancy forms). Overlays ensure the cashier doesn't lose the context of the underlying sale.

4. FULL SCREEN INVENTORY

Authentication & Setup

1. Splash / Loading Screen: Syncing cache, loading the latest menu and deals.
2. Login Screen: Username + PIN/Password keypad. Optimized for fast entry on shared terminals.

Sales Module (Core)

3. Main POS Layout (Sales):
 - *Zones:* Left Sidebar, Center Product Grid (Categories at top as pills/tabs), Right Cart Panel.
 - *Actions:* Tap item to add, select category, tap item in cart to edit.
4. Item Options / Notes Modal:
 - *Purpose:* Add "No salad", change quantity, confirm.
5. Payment Modal:
 - *Zones:* Total due, Payment Methods (Cash, Card, Staff, Other), Quick Cash keys (e.g., 10 DT, 20 DT, 50 DT, Exact).
 - *Actions:* Proceed to validation, trigger printer.
6. QR Code Scan Mode:
 - *Purpose:* Full screen or large panel prompting to scan a customer's mobile order QR.
 - *Actions:* Fetch order data, populate cart for review, proceed to standard payment.
7. Order Cancellation Modal:
 - *Purpose:* Prompt for mandatory cancellation reason. If unauthorized, prompts for Manager PIN.

Operations & Utilities

8. Today's Sales (Ventes du Jour):
 - *Zones:* Table of tickets (time, ticket #, total, payment mode, status). Top/Bottom Recap (Total CA, Cash, Card).
 - *Actions:* Select ticket to view details, Initiate Cancel.
9. Manual Cash Drawer Modal:
 - *Purpose:* Snapshot of current CA/Cash/Card. Mandatory reason text field to trigger opening.

Stock & Purchases

10. Stock List Screen:

- *Purpose:* Table of raw materials (qty, threshold, price). Visual alerts (red) for low stock.

11. Purchases List Screen:

- *Purpose:* History of supplier purchases.

12. New Purchase Flow (Split Screen/Modal):

- *Steps:* Add raw materials, set qty & purchase price.
- *Actions:* Upload invoice (Camera capture), final Validation.

Closures (End of Session / Day)

13. Session Closure Modal/Screen:

- *Zones:* Expected amounts vs Actual amounts inputs (Cash, Card, Fond de caisse).
- *Actions:* Mandatory note if discrepancy exists, TPE Photo Upload, Submit.

14. End-of-Day Stock Verification Screen:

- *Zones:* List of tracked items with theoretical stock. Inputs for real weighed/counted stock.
- *Actions:* Mandatory note for discrepancies, Finalize Day.

System States

15. Permission Denied / Override Modal:

- *Purpose:* Prompts for Manager PIN when a cashier attempts an unauthorized action (e.g., Discount).

5. MAIN USER FLOWS

Flow 1: Direct Cashier Sale

- *Trigger:* Customer walks up to the counter.
- *Steps:* Cashier taps Category -> Taps Product(s). Item appears in persistent cart.
- *Step 2:* Cashier taps "Pay" in cart.
- *Step 3:* Payment modal opens. Selects "Cash". Enters amount given.
- *Success:* System records sale, drawer opens, change to give is displayed, tickets print (client & kitchen), KDS updated, returns to empty POS screen.

Flow 2: Mobile QR Order Flow

- *Trigger:* Customer shows QR code from mobile app.
- *Steps:* Cashier navigates to "QR Scan" tab (or uses a physical barcode scanner).
- *Step 2:* App fetches order via API. Items populate the cart immediately.
- *Step 3:* Cashier verifies with the customer, taps "Pay", completes payment standard flow.
- *Success:* QR expires immediately upon confirmation.

Flow 3: Order Cancellation

- *Trigger:* A mistake was made, or the customer requests a refund.
- *Steps:* Cashier goes to "Today's Sales" -> selects Ticket -> Taps "Cancel Order".
- *Decisions:* System checks privileges. If cashier lacks rights, prompts for Manager PIN.
- *Validations:* Mandatory reason text field appears.

- *Success:* Ticket status changes to "Annulée", totals re-calculated, stock unaffected.

Flow 4: Manual Cash Drawer Opening

- *Trigger:* Need to give change for a high bill outside a sale.
- *Steps:* Tap "Open Drawer" utility icon.
- *Validations:* Modal shows a current financial snapshot. Disables the "Open" button until a Reason is typed.
- *Success:* Opens drawer, logs the event.

Flow 5: Session Closure Flow (Clôture Caisse)

- *Trigger:* Cashier shift ends.
- *Steps:* Tap "Clôturer caisse".
- *Decisions:* View expected totals. Count drawer and enter real Cash, real TPE total.
- *Validations:* If `Real != Expected`, note input becomes mandatory. TPE photo upload is mandatory.
- *Success:* Session locked, data sent to Admin for final approval.

Flow 6: Purchase Entry Flow

- *Trigger:* Delivery truck arrives with raw materials.
- *Steps:* Manager goes to "Purchases" -> "New Purchase". Adds items, qtys, prices.
- *Validations:* Taps "Upload Invoice", takes a photo with the POS terminal camera.
- *Success:* Stock is incremented automatically, invoice saved.

6. SCREEN GROUPING RECOMMENDATION

Group the app into these primary modules, accessible exclusively from the persistent sidebar:

1. Sales (Caisse) (Default view: Catalog + Cart)
2. Scan Order (QR) (Dedicated large scanning UI)
3. History (Ventes du jour) (Tickets, details, and cancellations)
4. Inventory (Stock & Achats) (Sub-tabs for Stock Levels vs Purchases)
5. Closures (Clôtures) (Session and End-of-Day forms)
6. Utilities (Manual Drawer, Switch User)

7. HIGH-PRIORITY SCREEN FOR STITCH

When starting the design system, prioritize in this exact order:

Batch 1: Core Selling (Must Design First)

1. Main POS Layout (Sidebar, Categories, Product Grid, Cart).
2. Payment Modal (Cash, Card, Numpad interface).
3. Item Options / Note Modal (Quantity, Custom requests).
4. Login / User Switch Screen.

Batch 2: Daily Operations 5. Today's Sales List & Ticket Detail Table. 6. Order Cancellation Modal + Manager Override popup. 7. QR Code Scan Screen (and Cart Import preview state). 8. Manual Cash Drawer Opening Modal.

Batch 3: Management & Stock 9. Session Closure Form (with TPE photo capture and Discrepancy logic). 10. End-of-Day Stock Verification Form. 11. Stock List & Purchases Entry (with Invoice photo capture).

8. UX RULES FOR A POS TERMINAL APP

- Landscape-First: Terminals are wide. Never design for portrait. Real estate must be optimized horizontally.
 - Persistent Cart Visibility: The ticket/cart must NEVER be hidden during the sales flow. It anchors the cashier's focus.
 - Large Touch Targets: Minimum 48x48dp for buttons. Critical buttons (Pay, Items) should be massive (64dp+).
 - Fast Typology: Keypads should be classic numpads layout, large and easy to tap by muscle memory alone.
 - Color for Status: `<Paid>` in solid Green, `<Annulée>` in solid Red, `<Draft/Pending>` in neutral gray.
 - Safe Placement of Destructive Actions: Actions like "Cancel Order" or "Clear Cart" must be far away from "Pay" and always require confirmation dialogs.
 - Error Handling & Loading: Empty states must be descriptive. Network loading states must freeze inputs to prevent double-charging.
 - Photo Upload Handling: Triggering an upload should open a full-screen camera overlay with bounding box brackets (to frame invoices/TPE tickets properly).
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9. DESIGN SYSTEM / LAYOUT GUIDELINES FOR STITCH

- Density Level: Medium-high density for the product grid (image + name + price) to minimize scrolling, but low density (high spacing) for the Cart to ensure flawless readability of the order items.
 - Spacing Guidelines: Multiples of 8px. Maintain large padding (24px - 32px) around primary action buttons to prevent mis-clicks.
 - Typography: Sans-serif, highly legible and bold (e.g., Inter, Roboto). Prices must be bold and monospaced for alignment in the cart.
 - Visual Presentation:
 - *Categories*: Scrollable pill tabs at the top of the grid.
 - *Financial Summaries*: Large, bold numbers. Distinguish between expected vs real clearly.
 - *Data*: Use Tables for "Today's Sales" and "Stock" (easy to scan rows). Use Cards for Products.
 - Modals: Must heavily dim the background (opacity 60%+) and lock background scrolling.
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10. BUSINESS RULES THAT IMPACT THE UI

Stitch must encode these logic rules directly into the UI states:

- Mandatory Notes Logic: During closures or drawer openings, if there is a discrepancy. The UI must dynamically evaluate `abs(Expected - Real) > 0` to toggle the required state of the Note text field from disabled/optional to mandatory.
- Submit Button Blocking: If a note is mandatory, the Submit button remains disabled until at least 5 characters are typed.
- Permission Fencing UI: Tapping a restricted button (e.g., Apply Discount) does not show a flat "Error". It instantly shows a "PIN Override" modal for a manager to authorize that specific action.
- Zero Stock Impact Rule: The Order Cancellation UI should include a subtle info tag indicating "Stock not impacted".
- QR Expiration State: The POS UI must handle cases where a scanned QR is > 5 minutes old by showing a distinct "Expired Order" rejection state, without populating the cart.
- Closure Checkpoints: The "Finalize Day" button in session closure is hard-disabled until the `TPE Photo` block contains an image.

11. WHAT BELONGS IN POS APP VS WHAT SHOULD STAY OUT

To prevent the POS from becoming cluttered and confusing, strict boundaries must be kept.
IN the POS App (Operational layer only):

- Fast Cashier Sales & Payment
- QR Mobile Order Scanning
- Cashier Day/Session Closures
- Purchase Entry + Invoice Capture (handled by on-site managers)
- Stock Level Consultation + Daily manual weighing

OUT of the POS App (Do NOT Include):

- System Setup (Taxes, Global payment methods, point conversion rates) -> Admin
- Menu Editing (Creating products, changing prices, editing recipes/BOM) -> Admin
- Creating/Editing Users and defining roles -> Admin
- Deep historical dashboards spanning months/years -> Admin
- KDS Drag/Drop flow (To Do / Doing / Done) -> KDS Tablet App

12. FINAL RECOMMENDED POS APP STRUCTURE

App Sections (Sidebar):

1. Sale / Caisse
2. Mobile Scan
3. Activity (Journals, Today's Sales)
4. Inventory (Stock & Achats)
5. End of Shift (Clôtures)

Screen List:

- Login Pinpad
- Main Sale Dashboard (Grid + Cart)

- Payment Terminal Modal
- Options/Note Modal
- Scanner View
- Today's Sales Table
- Order Details Sheet
- Cancel/Refund Modal
- Hardware Trigger (Open Drawer Modal)
- Stock Monitor List
- New Purchase Split-view
- End of Shift Audit Form
- EOD Stock Weighing Form

First Design Batch for Stitch: Layout frame (Sidebar + Split pane), Product Grid, Persistent Cart, Keypad, and Payment Modal.